# Non-celiac gluten sensitivity: literature review.

BACKGROUND: A significant percentage of the general population report problems caused by wheat and/or gluten ingestion, even though they do not have celiac disease (CD) or wheat allergy (WA), because they test negative both for CD-specific serology and histopathology and for immunoglobulin E (IgE)-mediated assays. Most patients report both gastrointestinal and nongastrointestinal symptoms, and all report improvement of symptoms on a gluten-free diet. This clinical condition has been named non-celiac gluten sensitivity (NCGS). AIM: We attempt to define the current pathogenic, clinical, and diagnostic criteria of this "new" disease, to provide a practical view that might be useful to evaluate, diagnose, and manage NCGS patients. METHODS: We reviewed the international literature through PubMed and Medline, using the search terms "wheat (hyper)sensitivity," "wheat allergy," "wheat intolerance," "gluten (hyper)sensitivity," and "gluten intolerance," and we discuss current knowledge about NCGS. RESULTS: It has been demonstrated that patients suffering from NCGS are a heterogeneous group, composed of several subgroups, each characterized by different pathogenesis, clinical history, and, probably, clinical course. NCGS diagnosis can be reached only by excluding CD and WA. Recent evidence shows that a personal history of food allergy in infancy, coexistent atopy, positive for immunoglobulin G (IgG) antigliadin antibodies and flow cytometric basophil activation test, with wheat and duodenal and/or ileum-colon intraepithelial and lamina propria eosinophil counts, could be useful to identify NCGS patients. CONCLUSIONS: Future research should aim to identify reliable biomarkers for NCGS diagnosis and to better define the different NCGS subgroups. Key teaching points: • Most patients report both gastrointestinal and nongastrointestinal symptoms, and all agree that there is an improvement of symptoms on a gluten-free diet. • NCGS diagnosis can be reached only by excluding celiac disease and wheat allergy. • Patients suffering from NCGS are a heterogeneous group, composed of several subgroups, each characterized by different pathogenesis, clinical history, and, probably, clinical course. • A personal history of food allergy in infancy, coexistent atopy, positive IgG antigliadin antibodies (AGA) and flow cytometric basophil activation test, with wheat and duodenal and/or ileum-colon intraepithelial and lamina propria eosinophil counts, could be useful to identify NCGS patients. • Future research should aim to identify reliable biomarkers for NCGS diagnosis and to better define the different NCGS subgroup.